Intro to Data Science - HW 7

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```
# Enter your name here: Mars Schrag
```

USE OF ARTIFICIAL INTELLIGENCE: While you are allowed to use generative AI as a learning tool for this assignment (e.g., to help with brainstorming, to see how to code a certain concept, to help identify flaws in reasoning, or to spot confusing or underdeveloped code segments), it must not be used to input homework questions, problems, or instructions directly, and you may not copy and paste any AI-generated content into your work. By submitting this assignment, you confirm that all the work is your own.

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```
# 4. I did this homework with help from ChatGPT or another generative AI tool,
# but I followed the homework guidelines.
```

In our last assignment, we explored data visualization in R using the **ggplot2** package. This homework continues to use ggplot2, but this time, with **maps**. Additionally, we will merge datasets using the built-in **merge()** function, which provides a similar capability to a JOIN in SQL (don't worry if you do not know SQL). Many analytical strategies require joining data from different sources based on a key - a field that two datasets have in common.

Step 1: Load the data

A. Some people may struggle to eat a healthy diet due to a lack of access to supermarkets, grocery stores, or other sources of healthy and affordable food. This limited availability to food is commonly referred to as **food insecurity**. However, as it is frequently linked to deep-rooted systemic injustices, many agree that the term "food apartheid" is more fitting.

There are numerous methods for measuring food store access and determining which areas are low-access neighborhoods with few healthy food options. In this homework assignment, we will focus on accessibility to sources of healthy food, as measured by the distance to a store in a given area.

The dataset available at https://data-science-intro.s3.us-east-2.amazonaws.com/FoodInsecurity.csv includes a variable labeled LAPOP1_10 that indicates the count of individuals residing more than 1 mile (for urban areas) or 10 miles (for rural areas) away from a supermarket.

To analyze the data, you can read it into a dataframe named **insecurity_df** and visually inspect its contents.

Please note in a comment what geographic area an observation (a row in the dataframe) represents - whether it pertains to a single city, a state, or another region.

Finally, use a command to display the number of observations and variables in insecurity_df.

```
insecurity_df <- read.csv("https://data-science-intro.s3.us-east-2.amazonaws.com/FoodInsecurity.csv")
```

```r
# Comment: A row represents a county in a specified state
```

```r
nrow(insecurity_df)
```

```
## [1] 3142
```

```r
ncol(insecurity_df)
```

```
## [1] 9
```

B. Calculate the average of MedianFamilyIncome in insecurity_df. Explain in a comment why the mean() function might not work and how to fix it.

```r
mean(as.numeric(insecurity_df$MedianFamilyIncome), na.rm = TRUE)
```

```
## Warning in mean(as.numeric(insecurity_df$MedianFamilyIncome), na.rm = TRUE):
## NAs introduced by coercion
```

```
## [1] 63973.97
```

```r
# you have to make it numeric and take out na's
```

C. What is the population of the most populous county in the dataframe? Which state is it in?

```r
max(insecurity_df$Pop2010)
```

```
## [1] 9818605
```

```r
which.max(insecurity_df$Pop2010)
```

```
## [1] 205
```

```r
insecurity_df[205, ]
```

```
##          State             County Pop2010 LAPOP1_10 AveragePovertyRate
## 205 California Los Angeles County 9818605  641340.3        16.24263362
##     MedianFamilyIncome Largest_city              city_state abbr
## 205        79980.20418  Los Angeles Los Angeles, California   CA
```

```r
# 9818605, california
```

D. Without a reference point, it is difficult to understand the significance of the values in **LAPOP1_10** (remember, this is the variable that shows the number of people living too far from a store and thus considered at risk of food insecurity). Create a new column in insecurity_df called **insecurity_ratio** that represents the ratio of LAPOP1_10 to Pop2010. Explain what it means in a comment.

```r
insecurity_df$insecurity_ratio <- insecurity_df$LAPOP1_10/insecurity_df$Pop2010
```

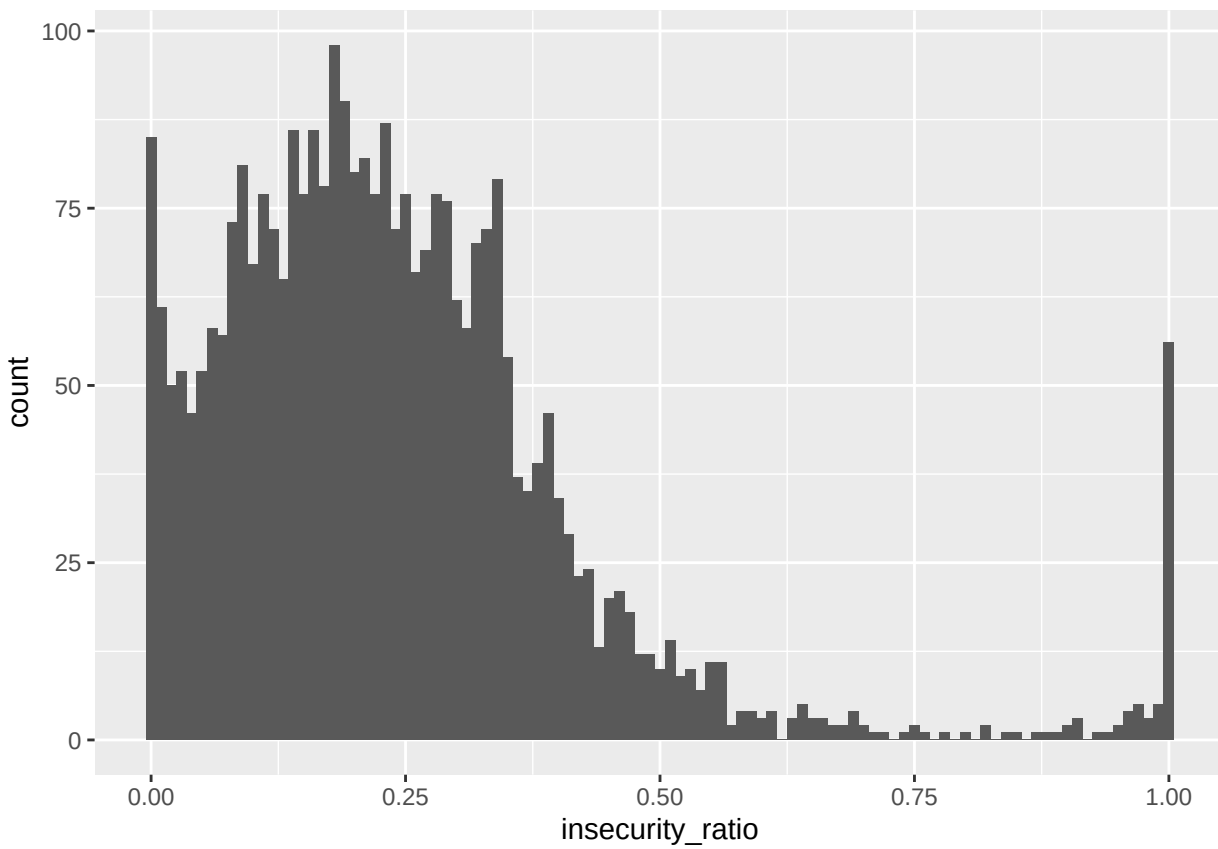
E. Provide descriptive statistics for the insecurity ratio (min, max, mean, and median) using a single function. Create a histogram with ggplot2 to visualize the distribution.

```r
summary(insecurity_df$insecurity_ratio)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.0000  0.1193  0.2147  0.2439  0.3237  1.0000
```

```r
library(ggplot2)

ggplot(data = insecurity_df, aes(x = insecurity_ratio)) + geom_histogram(binwidth = 0.01)
```



Step 2: Merge the food apartheid data with the city data

A. Read the JSON file at https://intro-datascience.s3.us-east-2.amazonaws.com/cities.json into a dataframe called **population_df**.

Use a function to display the variable data types in population_df and add comments explaining what each variable contains.

```r
library(jsonlite)

population_df <- fromJSON("https://intro-datascience.s3.us-east-2.amazonaws.com/cities.json")

str(population_df)
```

```
## 'data.frame':    1000 obs. of  7 variables:
##  $ city                    : chr  "New York" "Los Angeles" "Chicago" "Houston" ...
##  $ growth_from_2000_to_2013: chr  "4.8%" "4.8%" "-6.1%" "11.0%" ...
##  $ latitude                : num  40.7 34.1 41.9 29.8 40 ...
##  $ longitude               : num  -74 -118.2 -87.6 -95.4 -75.2 ...
##  $ population              : chr  "8405837" "3884307" "2718782" "2195914" ...
##  $ rank                    : chr  "1" "2" "3" "4" ...
##  $ state                   : chr  "New York" "California" "Illinois" "Texas" ...
```

```r
# city are cities growth shows percentage of population growth latitude is
# location of city longitude is also location of city population is current
# population in number rank is how big population is in comparison to other
# cities state is what state the city is in
```

B. To merge insecurity_df and pop_df, we need a common column. We'll use the **city_state variable** in insecurity_df, which we'll create by joining the city and state columns in population_df. Explain the code provided:

```r
population_df$city_state <- paste0(population_df$city, ", ", population_df$state)

# Comment: this makes one column with both the city and the state
```

C. Merge the two dataframes using the city_state column from both dataframes and store the resulting dataframe in **merged_df**.

```r
merged_df <- merge(insecurity_df, population_df, by = "city_state")
```

D. Display the dimensions (number of observations and variables) of merged_df and comment on the number of variables.

```r
dim(merged_df)
```

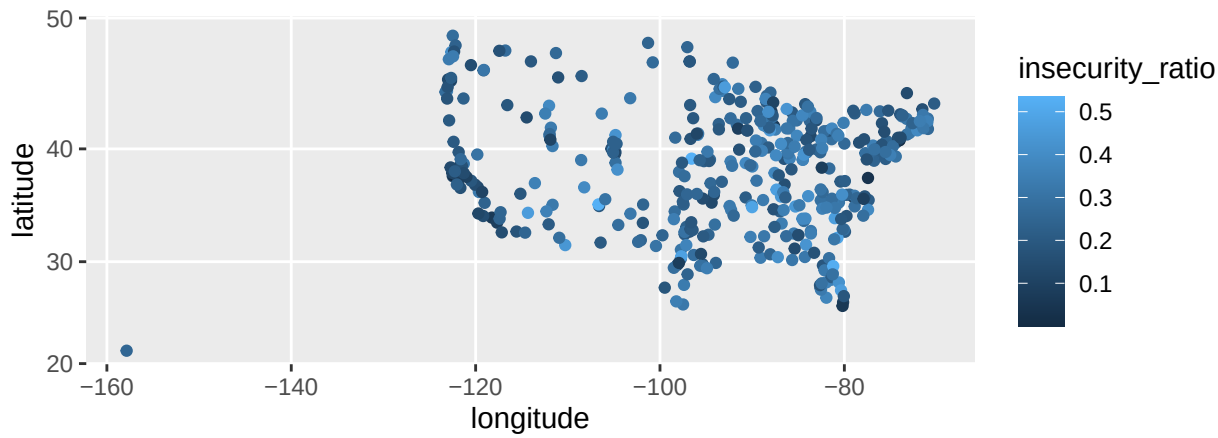
```
## [1] 423  17
```

```r
# 17 variables
```

Step 3: Visualize the data

A. Plot points for each city in merged_df on a map of the United States. Use color to represent the insecurity ratio.

```r
ggplot(merged_df) + aes(longitude, latitude) + geom_point(aes(color = insecurity_ratio)) +
    coord_map()
```



B. The map you just created is probably not perfect; comment briefly on what could be improved.

```r
# Comment: i cant figure out how to make the map in the background
```

Step 4: Group by State

A. Create a dataframe of state-by-state average **insecurity_ratio** using **group_by()** and **summarize()**. Save the result in **state_insecurity_df**.

```r
# state_insecurity_df$avg_insecurity_ratio <- mean(merged_df$insecurity_ratio)

# state_insecurity_df <- group_by(State) summarize(avg_insecurity_ratio =
# mean(insecurity_ratio, na.rm = TRUE))

# i dont know why group by and summarize wont work
```

B. Identify the most and least food-insecure states in state_insecurity_df and their food insecurity ratios. Show the code you used to determine them.

```
# which.min(state_insecurity_df$avg_insecurity_ratio)
# which.max(state_insecurity_df$avg_insecurity_ratio) my code is broken, all
# the averages are the same because group by didnt work
```

Step 5: Create a map of the United States, with the color of each state representing the average food insecurity ratio in state_insecurity_df

A. Ensure you correctly set the map limits and use coord_map appropriately, and make sure that lighter shades represent lower food insecurity.

```
# ggplot(data=state_insecurity_df, aes(x = longitude, y = latitude, fill =
# avg_insecurity_ratio)) + geom_polygon(color = 'white') + coord_map() my df is
# broken so it does not work
```

Step 6: Data Analysis

A. Explain in a comment what insights you can glean about food insecurity from the maps you created in this homework.

```
# Comment: The second map is broken, but I imagine there is a trend where food
# insecurity is higher in places where there are less large cities.
```